Chandan Kumar

Full-Stack Engineer | AI/ML Developer | Cloud Infrastructure Specialist

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Professional Summary

Innovative full-stack engineer with cross-domain experience across AI/ML, distributed systems and cloud infrastructure. Proven record building scalable cloud applications, on-device large-language-model inference engines and enterprise-grade systems. Proficient in eight-plus programming languages including **Python, JavaScript, Swift, Rust, C++, Go and SQL**. Skilled in serverless architectures, edge computing and modern iOS development. MBA candidate in Business Analytics who combines business insight with technical expertise to deliver high-impact solutions.

Technical Skills

Programming Languages

Python, JavaScript, Swift, Rust, C++, Go, R, Shell/Bash, HTML/CSS, SQL

Frameworks & Tools

- **AI/ML:** TensorFlow, PyTorch, scikit-learn, pandas, NumPy, Neo4j
- **Web & App Development:** Node.js, SwiftUI, UIKit, React/Next.js, HTML5/CSS3
- **DevOps & Infrastructure:** Git, Docker, Kubernetes, Terraform, CI/CD pipelines, Cloudflare Workers, Firebase, Wrangler CLI

Cloud & Infrastructure

Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), Cloudflare R2 & D1, Serverless Architecture, Edge Computing, NPU/TPU optimisation

Databases

SQL (MySQL/PostgreSQL), Graph DB (Neo4j), Key-Value Stores (Redis/Cloudflare KV), D1 relational DB

AI/ML & Data Science

Large-language-model (LLM) inference and optimisation, machine-learning algorithms (clustering, classification, regression), data visualisation (Tableau, Power BI), game theory and mechanism design frameworks

Business & Analytics

KPI tracking, data visualisation, dashboard creation, UI/UX design (Figma), product strategy

Experience

41Dayta – Founder & Product Lead | London, UK - Jan 2025 – Present

- Launched **41Dayta**, a privacy-first CRM platform for solopreneurs and small teams; grew to **200+ beta sign-ups**, integrating secure data storage and analytics.
- Built **Moomo.dev**, a multilingual large-language-model platform enabling real-time conversations in any language. Developed **AI-powered Word and PPT editors** that generate research papers and presentations from prompts.
- Launched **Moomo.News**, a dynamic AI-curated news portal delivering personalised updates.
- Led end-to-end product strategy and UI/UX design; optimised performance to achieve **30% faster load times** than comparable solutions while reducing database costs by **50%**.
- Managed cross-functional team; coordinated engineering and marketing to deliver MVP, gather user feedback and iterate quickly.

On-Device LLM Inference Engine – InSystem-Compute | Enterprise AI Infrastructure -Nov 2025 – Present

- Designed and implemented a **hardware-agnostic framework** supporting Rust, C++, Go and Python for deploying large language models on edge devices and distributed compute environments.
- Built a hardware abstraction layer to leverage **CPU, GPU, NPU and TPU acceleration**, reducing latency and cloud dependency for enterprise LLM workloads.
- Architected multi-language support with optimised inference pipelines, resulting in significant performance improvements across edge devices.
- Adopted serverless and distributed computing patterns; integrated CI/CD pipelines and robust monitoring to ensure production readiness and commercial licensing.

Movo – Contact Database & File Storage | Cloud-Native Application - Dec 2025 – Present

- Built and deployed **Movo**, a globally accessible serverless application for contact management and file storage using Cloudflare Workers and R2 storage.
- Implemented RESTful APIs with nine endpoints supporting CRUD operations, file upload/download and health monitoring.
- Designed relational database schema (files, contacts, file_contacts) on D1 DB and integrated KV namespace for **sub-millisecond alias lookups** and caching.
- Developed multipart upload system with metadata tracking and deduplication; optimised cost by reducing duplicate writes by **50%** and cutting API expenses by **60–80%**.
- Ensured zero-maintenance scaling with serverless architecture and implemented comprehensive monitoring and performance metrics.

CampusLive – iOS Developer | Dec 2025 – Present

- Developed a native iOS application for real-time campus engagement using **Swift** and modern **UIKit/SwiftUI** architecture.
- Integrated JavaScript bridge to support cross-platform functionality and web services.
- Created automation scripts for continuous integration and deployment; implemented live update features and push notifications.
- Enhanced user engagement by designing intuitive UI/UX and implementing real-time communication features.

AI Debater – Game Theory & Mechanism Design | Research Project - Nov 2020

- Developed an AI system applying **game-theory and mechanism-design principles** to strategic debating scenarios.
- Implemented algorithms to maximise utility in competitive debate; built argumentation frameworks using a **Neo4j graph database** for relationship mapping.
- Researched **Nash equilibrium** and strategic decision-making; integrated a Python-based AI to analyse and generate debate strategies.

Polls – Open-Source Voting & Survey Platform | Sep 2020

- Designed a **JavaScript-based polling application** with an intuitive user interface and responsive design.
- Deployed the application on GitHub Pages; managed contributions under the **MIT License**.
- Implemented features such as real-time results, accessible polling workflows and data export.

Additional Projects

- **Airlines Customer Clustering Analysis** (Data Science/Machine Learning): Conducted customer segmentation for East West Airlines using hierarchical and k-means clustering on **3,999 passenger records**, delivering targeted marketing strategies for four distinct customer segments and improving engagement.
- **Moomoo AI Assistant:** Built multiple iterations of voice-enabled AI assistants and chat interfaces using **Swift, JavaScript and Firebase**; achieved **50– 80% performance improvements** and cost reductions by optimising API usage and storage; implemented speech-to-text, multilingual support and robust user authentication.
- **TechTribe SEO & Paid Search Optimisation** (Digital Marketing/Data Analysis): Performed comprehensive SEO and paid search analysis for a large e-commerce platform; analysed **167 K monthly clicks** and **$755 K ad spend**, identified **77% keyword cannibalisation** and proposed strategies yielding **94% cost reduction** while maintaining key traffic metrics.

Education

Master of Business Administration (MBA) in Business Analytics, Illinois Institute of Technology, Chicago, IL - *Expected Dec 2025* | **GPA:** 3.3

Bachelor of Business Administration in International Business, Amity University, Noida, India - *Jun 2024* | **GPA:** 3.8

Exchange Programs: Adelphi University, New York - *May 2023*; Birkbeck University, London - *Dec 2022*

Honors & Awards

- **Best Student Award**, Birkbeck University (London), presented by Graham Bolton - *Mar 2023*
- **Gold Medal in Marathon**, Amity Education Group Championship - *May 2023*
- **Alpha Sigma Phi (ΑΣΦ) Member**, Illinois Institute of Technology - *Jun 2025*

Professional Affiliations & Open-Source Contributions

- Active contributor to multiple open-source projects; maintain GitHub repositories focusing on AI/ML, distributed systems, mobile apps and data analytics.
- Member of Illinois Institute of Technology's **Alpha Sigma Phi** fraternity and participant in campus technology initiatives.

Additional Information

- **Certificates & Training:** list relevant certifications such as AWS Certified Solutions Architect, Google Professional Cloud Architect or Machine Learning specialisations if applicable.
- **Languages:** Fluent in English; working proficiency in Hindi (add others as appropriate).
- **Interests:** Edge AI, serverless architectures, game theory, digital product design and community open-source development.